

Card 3: Secret Code Lab

The Hackles Kit

A free kit by Competence Collective · competencecollective.org · October 2026

Hackles Lab · Teacher card · Grades 3-8 · secret-code-lab/index.html

Learning goal. A cipher scrambles a message and a key unlocks it. A shift cipher has only 25 keys, so real encryption needs enormous keys.

SPECIMEN LABEL	
10 MIN	Shift wheel tab: set key 3 and watch PAUSE become SDXVH. Try Decode. Solve codes 1 and 2 in Crack it.
30 MIN	Add the Mixed alphabet tab (classroom key, then your own). Solve all 7 codes. After a shift code, open "all 25 keys" and ask why a computer finishes instantly. Discuss sharing a key safely: in person, never in the same note.

Screens. Wheel (outer ring is the message, inner ring is the code, key is the turn) with a letter map table. Mixed alphabet (the scrambled alphabet is the key; 403 septillion keys). Crack it (7 puzzles, no timer, less help as you go).

No free typing. Messages come from a list or word chips, so no personal details can be entered. Every ciphertext is computed in code from the plaintext, and a test checks the kit's published ciphers against the math.

Teach. Link to Passphrase Lab: 25 keys against 46,656 three-word passphrases. Encryption protects a message on its way; it does not show the other person is safe. A shift cipher is a toy.

Standards (approximate; verify). CSTA 1B-NI-05 and middle-grades encryption; ISTE 2d; CASEL Self-Management.

Accessibility. Wheel has a text alternative; keyboard and arrow-key tabs; polite results; Spanish puzzles follow the glossary (no enye, no accents) and are a draft.

Safety. If a child mentions a secret they were told to keep from adults, pause and follow your disclosure steps.

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